MH2500 Probability and Introduction to Statistics

Slides for Week 2 - 2 and Week 3 - 1

Topics: Probability, Conditional Probability and Bayes' Rule

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Experiment: Flip a coin until the first head appears

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Then

$$A' = \{H, TH\} \qquad A \cap B = \{TTH, TTTH\} \qquad A \cup B = \Omega$$

Probability

A formal definition of probability should agree with our naive ideas about probability:

1. $P(\emptyset) = 0$;
2. $P(\Omega) = 1$;
3. $P(A') = 1 - P(A)$;
4. $P(A \cup B) = P(A) + P(B)$, if A and B are mutually exclusive.

$\ldots\ldots$

Definition

A **probability measure** on Ω is a mapping P from subsets of Ω to the real numbers that satisfies the following conditions:

1. (nonnegativity) If $A \subseteq \Omega$, then $P(A) \geq 0$.
2. (unitarity) $P(\Omega) = 1$.
3. (countable additivity) If $A_1, A_2, \ldots$ are mutually disjoint, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

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We call these conditions axioms.

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This property implies *finite additivity*:

If $A_1, A_2, \ldots, A_n$ are mutually disjoint, then

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i).$$

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$$P(A) + P(A') = P(A \cup A') = P(\Omega) = 1,$$

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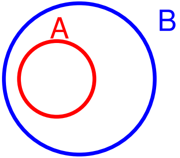
Hence

$$P(A) + P(A') = P(A \cup A') = P(\Omega) = 1,$$

and $P(A') = 1 - P(A)$.

Property C: If $A \subseteq B$, then $P(A) \leq P(B)$



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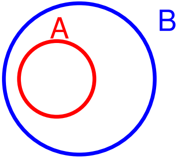

Proof: Since $A \subseteq B$, B as a disjoint union of A and $B \cap A'$, i.e., $B \backslash A$. Then

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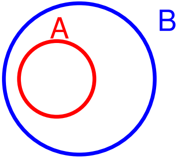

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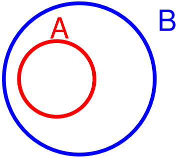

Proof: Since $A \subseteq B$, B as a disjoint union of A and $B \cap A'$, i.e., $B \backslash A$. Then

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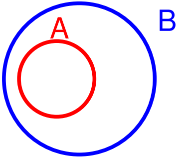

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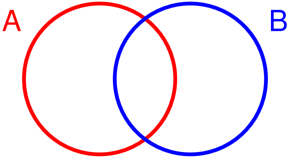
$$P(B) = P(A \cup (B \cap A')) = P(A) + P(B \cap A') \geq P(A) + 0 = P(A).$$

Property D: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

We call this the addition rule.



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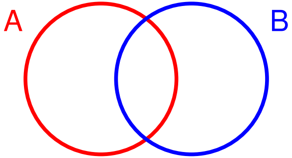

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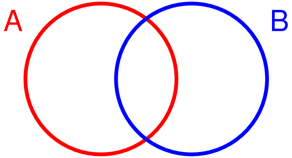

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As $A = \{THH, HTH, HHT, HHH\}$ and $B = \{TTT, TTH, HTT, HTH\}$,

$$P(A) = \frac{4}{8} = \frac{1}{2}, \qquad P(B) = \frac{4}{8} = \frac{1}{2}, \qquad P(A \cap B) = \frac{1}{8}.$$

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and $P(A \cup B) = \dfrac{7}{8}$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$= \frac{1}{2} + \frac{1}{2} - \frac{1}{8} = \frac{7}{8}$$

Computing Probabilities

For a finite sample space $\Omega = \{\omega_1, \omega_2, \ldots, \omega_N\}$, if for each i,

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then for any event A,

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In particular, if every ω_i has the same probability, then by $Np_i = p_1 + \cdots + p_N = 1$,

$$P(\omega_i) = \frac{1}{N},$$

and if $A \subseteq \Omega$ contains n elements, we have

$$P(A) = np_i = n \cdot \frac{1}{N} = \frac{n}{N} = \frac{\text{number of ways } A \text{ can occur}}{\text{total number of outcomes}}.$$

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We now assume that $n \leq 365$. As A' is the event that no two people have a common birthday, i.e., everyone have a different birthday, we have

$$P(A') = \frac{365 \times 364 \times \cdots \times (365 - n + 1)}{365^n},$$

$$P(A) = 1 - P(A') = 1 - \frac{365 \times 364 \times \cdots \times (365 - n + 1)}{365^n}.$$

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When $n = 70$, $P(A) = 0.9992$.

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$$P(A') = \frac{364^n}{365^n}$$

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Then $\left(\dfrac{364}{365}\right)^n = \dfrac{1}{2} \implies n = \dfrac{\ln(1/2)}{\ln(364/365)} \approx 252.65$

Property E: For an increasing sequence of events $A_1 \subseteq A_2 \subseteq \cdots$, $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{n\to\infty} P(A_n)$

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Proof. Let $B_1 = A_1$ and $B_2 = A_2 \backslash A_1$, $B_3 = A_3 \backslash A_2$, $\cdots$, $B_n = A_n \backslash A_{n-1}$, $\cdots$.

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Proof. Let $B_1 = A_1$ and $B_2 = A_2 \backslash A_1$, $B_3 = A_3 \backslash A_2$, $\cdots$, $B_n = A_n \backslash A_{n-1}$, $\cdots$.

Then for any $n \geq 1$,

$$A_n = \bigcup_{i=1}^{n} B_n,$$

and

$$
\begin{aligned}
\bigcup_{i=1}^{\infty} A_i &= \lim_{n\to\infty} \bigcup_{i=1}^{n} A_i \quad \text{(by definition of infinite union)}\\
&= \lim_{n\to\infty} A_n \quad (\text{as } A_1 \subseteq A_2 \subseteq \cdots \subseteq A_n)\\
&= \lim_{n\to\infty} \bigcup_{i=1}^{n} B_i \quad \left(\text{we had } A_n = \textstyle\bigcup_{i=1}^{n} B_n \text{ just now}\right)\\
&= \bigcup_{i=1}^{\infty} B_i \quad \text{(by definition of infinite union again).}
\end{aligned}
$$

This gives

$$
P\left(\bigcup_{i=1}^{\infty} A_i\right) = P\left(\bigcup_{i=1}^{\infty} B_i\right) = \sum_{i=1}^{\infty} P(B_i) = \lim_{n\to\infty} \sum_{i=1}^{n} P(B_i) = \lim_{n\to\infty} P(A_n).
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What does $P(A) = 0$ mean? How about $P(A) = 1$?

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- Zero Probability / Almost Never Happens does not mean Impossibility.
- Almost Surely does not mean Certainty.

Inclusion-Exclusion Principle

For events A, B, C,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C)$$
$$+P(A \cap B \cap C).$$

Example

A total of 36 members of a club play tennis, 28 play squash, and 18 paly badminton. Furthermore, 22 of the members play both tennis and squash, 12 play both tennis and badminton, 9 play both squash and badminton, and 4 play all three sports.

How many members of this club play at least one of the three sports?

Let A be the set of members that plays tennis, B be the set that plays squash, and C be the set that plays badminton. Then we have

- $|A| = 36$, $|B| = 28$, $|C| = 18$,

  $|A \cap B| = 22$, $|A \cap C| = 12$, $|B \cap C| = 9$.

  $|A \cap B \cap C| = 4$.

Thus, by the inclusion-exclusion principle,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap c| + |A \cap B \cap C|$$

$$= 36 + 28 + 18 - 22 - 12 - 9 + 4 = 43.$$

Let 100 denote the number of members of the club, and introduce probability by assuming that a member of the club is randomly selected.

Then we have the following probabilities:

- $P(A) = \dfrac{36}{100}, P(B) = \dfrac{28}{100}, P(C) = \dfrac{18}{100},$

  $P(A \cap B) = \dfrac{22}{100}, P(A \cap C) = \dfrac{12}{100}, P(B \cap C) = \dfrac{9}{100},$

  $P(A \cap B \cap C) = \dfrac{4}{100}, P(A \cup B \cup C) = \dfrac{43}{100}.$

Conditional Probability

Definition

Let A and B be two events with $P(B) \neq 0$. The conditional probability of A given B is defined to be

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

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This induces a probability measure treating B as the sample space.

Example

A student is taking a one-hour-time-limit make-up examination.

- Suppose the probability that the student will finish the exam in *less than* x hours is $x/2$, for all $0 \le x \le 1$.

Then, given that the student is still working after 0.75 hour, what is the conditional probability that the full hour is used?

Solution: We use L_x to denote the event that student finishes the exam in less than x hours. Then

- L_1' is the event that the student uses the full hour and $P(L_1') = 1 - P(L_1) = 1 - 1/2 = 1/2$.
- $L_{0.75}'$ is the event that the student still works after 0.75 hour, and $P(L_{0.75}') = 1 - P(L_{0.75}) = 1 - 3/8 = 5/8$.

Thus,

$$P(L_1'|L_{0.75}') = \frac{P(L_1' \cap L_{0.75}')}{P(L_{0.75}')} = \frac{P(L_1')}{P(L_{0.75}')} = \frac{\frac{1}{2}}{\frac{5}{8}} = \frac{4}{5}.$$

Example

A fair coin is flipped twice. What is the conditional probability that both flips land on heads, given that

(a) the first flip lands on heads?

(b) at least one flip lands on heads?

Solution: We let

- A denote the event $\{HH\}$, exactly two heads;
- B denote the event $\{HT, HH\}$, the first flip is head;
- C denote the event $\{HH, HT, TH\}$, at least one head.

Then $\underline{\underline{P(A) = 1/4}}$, $\underline{\underline{P(B) = 1/2}}$ and $\underline{\underline{P(C) = 3/4}}$.

Also we have $P(A \cap B) = P(A) = 1/4$, and $P(A \cap C) = P(A) = 1/4$.

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Thus

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{1/2} = \frac{1}{2};$$

$$P(A|C) = \frac{P(A \cap C)}{P(C)} = \frac{1/4}{3/4} = \frac{1}{3}.$$

Multiplication Rule

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$$P(A_1 \cap A_2 \cap A_3) = P(A_3|A_1 \cap A_2)P(A_1 \cap A_2) = P(A_3|A_1 \cap A_2)P(A_2|A_1)P(A_1).$$

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We can generalize this to n many events, for arbitrary n.

Example

Suppose that an urn contains 8 red balls and 4 white balls.

- We draw 2 balls from the urn without replacement uniformly at random.

What is the probability that both balls drawn are red?

Solution: By counting:

- There are $\binom{12}{2}$ many ways to draw two balls from 12 balls.
- There are $\binom{8}{2}$ many ways to draw two red balls from 8 red balls.
- Thus, the probability that both balls drawn are red is $\frac{\binom{8}{2}}{\binom{12}{2}} = \frac{14}{33}$.

By conditioning: We let $A = \{RW, RR\}$, $B = \{RR, WR\}$. That is, A is the event that the first ball we draw is a red ball, and B is the event that the second ball we draw is a red ball. Then $A \cap B$ is the event that two ball we draw are both red.

- $P(A) = \frac{8}{12}$;
- $P(B|A) = \frac{7}{11}$.

Hence,

$$P(A \cap B) = P(A)P(B|A) = \frac{8}{12} \cdot \frac{7}{11} = \frac{14}{33}$$

Law of Total Probability

Let $B_1, B_2, \ldots B_n$ be such that $\bigcup_{i=1}^{n} B_i = \Omega$ and $B_i \cap B_j = \varnothing$ for $i \neq j$ with $P(B_i) > 0$ for all i. Then for any event A, $P(A) = \sum_{i=1}^{n} P(A|B_i)P(B_i)$.

Proof:

$$
\begin{aligned}
P(A) &= P(A \cap \Omega) = P\left(A \cap \bigcup_{i=1}^{n} B_i\right) && \left(\text{by } \bigcup_{i=1}^{n} B_i = \Omega\right) \\
&= P\left(\bigcup_{i=1}^{n} (A \cap B_i)\right) && (\text{by set identity}) \\
&= \sum_{i=1}^{n} P(A \cap B_i) && (\text{by mutually disjointness}) \\
&= \sum_{i=1}^{n} P(A|B_i)P(B_i) && (\text{by multiplication law})
\end{aligned}
$$

Example

An urn contains three red balls, two blue balls, and two white balls. Two balls are selected without replacement. What is the probability that the second ball selected is red?

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Solution: Let R_1 denote the first ball selected is red, and R_2 denote the second ball selected is red. Then

$$P(R_2) = P(R_2|R_1)P(R_1) + P(R_2|R_1')P(R_1')$$

$$= \frac{2}{6} \cdot \frac{3}{7} + \frac{3}{6} \cdot \frac{4}{7} = \frac{3}{7}$$

Bayes' Rule

Let A and $B_1, B_2, \ldots, B_n$ be events where the B_i's are disjoint, $\bigcup_{i=1}^{n} B_i = \Omega$, and $P(B_i) > 0$ for all i. Then

$$P(B_j|A) = \frac{P(A|B_j)P(B_j)}{\sum_{i=1}^{n} P(A|B_i)P(B_i)}.$$

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Proof: By definition of conditional probability,

$$P(B_j|A) = \frac{P(A \cap B_j)}{P(A)}.$$

The numerator is $P(A \cap B_j) = P(A|B_j)P(B_j)$. By law of total probability, the denominator is $P(A) = \sum_{i=1}^{n} P(A|B_i)P(B_i)$.

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Example: Guessing MC Questions

For a multiple-choice question, a student knows the answer with probability p and guesses with probability $1 - p$.

- Assume that a student who guesses at the answer will be correct with probability $1/m$, where m is the number of multiple-choice alternatives.

What is the conditional probability that a student knew the answer to a question given that he or she answered it correctly?

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- Assume that a student who guesses at the answer will be correct with probability $1/m$, where m is the number of multiple-choice alternatives.

What is the conditional probability that a student knew the answer to a question given that he or she answered it correctly?

Solution: Let C denote the event that the answer is correct and K that the student actually knows the answer.

What we want is $P(K|C)$.

$$P(K|C) = \frac{P(C|K)P(K)}{P(C|K)P(K) + P(C|K^c)P(K^c)}$$

$$= \frac{1 \cdot p}{1 \cdot p + \frac{1}{m} \cdot (1-p)}$$

$$= \frac{mp}{mp + (1-p)}$$

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When $m = 4$ and $p = 1/2$, $P(K|C) = 4/5$.